CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (Physics) Examination March 2018

PS909 Introduction to Nanotechnology

Date: 28/03/2018 Day: Wednesday   Time: 1.30 pm To 1.50 pm   Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. (one mark each). 10

1. Who coined the word Nanotechnology"?

(i) K E Drexler   (ii) Richard Feynman
(iii) Nario Taniguchi   (iv) Jeremy Ramsden

2. Nanoparticles are the examples of _____ structure.

(i) Zero dimensional   (ii) One dimensional

(iii) Two dimensional   (iv) None of the above

3. Which of the following forces is most important for the self-assembly?

(i) Electrostatic force   (ii) Gravitational force

(iii) Nuclear force   (iv) van der Waal force

4. How many carbon atoms are used to form Fullerene or bucky ball ?

(i) 20   (ii) 60

(iii) 75   (iv) 100

5. Which of the following does not relates to Quantum Dots (QD)?

(i) QD are small metal particles

(ii) QD are small semiconductor particles

(iii) QD is a single object with bound, discrete electronic states

(iv) QD tightly confine either electrons or electron holes

6. Understanding of Nanoscience is possible due to the development of __________.

(i) Quantum mechanics

(ii) Newtonian mechanics

(iii) Electrodynamics

(iv) Atomic Physics

7. The width of a typical DNA molecule is ____ nm.

(i) 1

(ii) 2

(iii) 5

(iv) 10

8. Which instrument was used to discover fullerene?

(i) Electron microscope

(ii) Magnetic resonance

(iii) Condensation technique

(iv) Mass spectrograph

9. Which ratio decides the efficiency of nanosubstances?

(i) Weight/volume

(ii) Surface area/volume

(iii) Volume/weight

(iv) Pressure/volume

10. Who had invented the famous 'Geodesic' dome structure?

(i) Eric Drexler

(ii) Buckminster Fuller

(iii) Richard Smalley

(iv) Faraday

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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (Physics) Examination March 2018

PS909 Introduction to Nanotechnology

Date: 28/03/2018 Day: Wednesday   Time: 1.50 pm To 3.30 pm   Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q – II Attempt All.

a) Explain the mechanism of binding of nanoparticles when delivered to biological cells. Why at nanoparticles of silver is toxic compared to its bulk form? **[4]**

OR

Write a note: (i) Graphene, (ii) carbon nanotubes and (iii) carbon nanoparticles.

b) Write a truth-table for NAND gate. Design a circuit that works as NAND gate. Also, draw its input-output relationships approaching the ideal step function. How this step function has changed the miniaturization function of memory elements? **[4]**

c) How the semiconductor nanoparticles may provide alternative solution to the current energy production problem? **[2]**

OR

By varying the radius of a particle, the optical absorption and emission wavelength can be tuned: Justify the statement.

SECTION - II

Q-III Attempt all.

a) Why and how will the quantum computer take over the present generation computer? Explain the function and role of spintronic NAND gate in quantum computers. **[4]**

b) What are the pros and cons of externally administrative (i.e. oral, intravenous, etc.) pharmaceutical drugs. **[3]**

OR

Write a note on Custom synthesis of drugs.

c) What is superparamagnetism? How ferromagnetic substance behaves as superparamagnetic at nano-scale? **[3]**

d) Write a note on giant magnetoresistance (GMR) with reference to the development of magnetic nanodevices. **[3]**

e) Why the probability of occurrence of impurities reduces with reduction in particle size? **[2]**